Chapter 4

Political participation

Where might we begin when studying the political consequences of this identity? We know that to belong to your neighbourhood is an identity more widely held than attachment to country, and that it is strongly driven by some of the fundamental ways in which we relate to each other and to our environment. It seems likely then that, far from being on the periphery of politics, or incidental to political behaviour, place attachment plays an integral role to how we engage in politics in democratic societies. We could look at several areas. We might consider how this identity drives party choice. We might also consider liberal-authoritarian attitudes, or attitudes towards local policies. But it is better to start with something more fundamental; as important as who or what we vote for is whether we vote at all. As important again is the realm of political engagement beyond voting: what causes us to display our allegiances publicly; to protest; to campaign; or to engage with the politicians we elect.

Studying participation is also one in which the conclusions we have already drawn from studying this identity are most pertinent. Political science provides us with diverse answers to questions: what causes us to engage in politics? What behaviours do we engage in when we do? Are we driven more by our own circumstances, or of those around us? While we know about how place, space and the physical environment influence participation, what we know far less about is the effect of how we relate to these places on participation. Social and environmental psychology and sociology indicate that attachment to place drives participatory behaviours (Davidson and Cotte 1989; Hays and Kogl 2007). A growing literature within political science studies the effect on political participation specifically (Borwein and Lucas 2021; Bühlmann 2012; Fitzgerald 2018; Lappie and Marschall 2018; Lin and Lunz Trujillo 2022; Wong 2010). Not only is the contextual environment key in influencing participation, but how we relate to it and the groups in it, as with Charles Mackay's famous adage that 'men... think in herds; it will be seen that they go mad in herds, while they only recover their senses slowly, and one by one' (Mackay 1852 page *XX*). Though the effect of social context on participation is well-studied, some of the omissions which I discussed in Chapter 3 are yet more significant here, in particular a neglect to study variation on small units—where attachment to geographic social groups and the factors which shape participation may be strongest—and a focus on small samples and cross-sectional data.

This chapter sets out to address three key questions. First, does being attached to place on a very local level cause us to engage more in politics? Secondly, is there heterogeneity between different activities? And finally, is there heterogeneity between areas? Specifically, is any potential relationship differentiated by contextual political congruence: whether an area votes like us. It begins by summarising the theory: looking at neighbourhood context and social-psychological explanations. To study this, I draw on observational panel data, matching to micro-level vote counts. I begin with Round 6 (2012) of the European Social Survey (ESS), then three waves of the UKHLS, which allows me to model on neighbourhoods. The UKHLS also allows me to test parts of the mechanism, and explore variation by the political makeup of the area based on real-world electoral microdata which I have collated.

Participation and attachment

We already know much about how many individual-level attributes—income, age, education, interest in politics and information—affect political participation. A literature nearly as varied is also concerned with the role of context and environment in shaping participation. Studies of 'neighbourhood effects' on voting behaviour date back to the middle of the last century (Butler and Stokes 1971; Cox 1969): an area that seeks to understand spatial variation in views and voting behaviour which cannot be explained by personal factors (Biggs and Knauss 2012; Blok and Meer 2018; Johnston et al. 2000; Johnston et al. 2005). V.O. Key's seminal midcentury work documented what he called the 'friends and neighbors' effect in the southern United States, in which, partly, voters would be more likely to support candidates if their close friends or neighbours supported them, even if they had held an initially different position (Key 1949). Subsequent studies confirmed the presence of this phenomenon, not just within the United States, but around the world (Arzheimer and Evans 2012; Arzheimer and Evans 2014; Gimpel et al. 2008; Górecki and Marsh 2014; Jankowski 2016; Lewis-Beck and Rice 1983; Tatalovich 1975). These can be contextual effects—that group-level factors, such as the character of the environment or the social makeup of the neighbourhood, influence individual behaviour—or compositional—that the behaviour of a group simply reflects the sum of behaviours of individuals within that group (Maxwell 2019; Wijk, Bolt, and Johnston 2019). Studies also attempt to explain local variation in turnout through local sociodemographic or economic factors, particularly inequality, the structural features of urban versus rural areas, size of electoral unit, and the behaviour of voters in relation to their neighbours (Lappie and Marschall 2018; Lin and Lunz Trujillo 2022; Zollinger 2024).

What we often ignore, however, is not the physical or social characteristics of place and their effect on participation, but the ways in which we relate emotionally to place and how this affects how we participate in politics. Political science has established group identity as a strong motivator of political behaviour. We know that attachment to social groups—ethnicity, age, gender, class—can influence political action, from engagement in politics to how we vote, and how we evaluate political issues and other partisans (Conover 1984; Cramer 2016; Huddy 2013; Huddy, Mason, and Aarøe 2015; Miller et al. 1981). This effect carries across both formal membership of such groups, and group consciousness, denoting both cognitive and affective attachment. Researchers are also interested in the role environment itself and context play in shaping political participation (Bartle, Birch, and Skirmuntt 2017; Cho and Rudolph 2008; Enos 2017; Jöst 2023; Zollinger 2024). We therefore know much about how context and the local environment shape participation; we also know much about the effect of social influence—including group behaviour, social observation, and peer networks within that environment on political behaviour, including participation, vote choice and donating (Gerber, Green, and Larimer 2008; McClendon 2014; Mckenzie 2004).

In the past decade, political science has come to study in greater depth the effect of our emotional attachment to that environment. Cara Wong (2010), in a prominent early work highlights how identification with community motivates attitudes towards fiscal policy, and local political engagement, focusing on capturing 'closeness' to community. Research which studies the effect of named municipal identities on participation is fairly typical of such studies. Borwein and Lucas (2021) use two large surveys of residents in the Canadian cities of Calgary and Ontario, finding a strong relationship between attachment to these cities and interest in local politics. Bühlmann (2012) finds attachment to Swiss municipalities has an effect on civic and political engagement. [1] Most of these studies find a positive relationship. Where they disagree on this effect of place attachment on participation is usually from the confounding effect of smaller electoral units—generally rural and more isolated—generate greater turnout

[1] Many of these study place attachment using proxy measures. See Borwein and Lucas (2021) and John et al. (2011) for exceptions. Approaches to measurement of place attachment in the literature are also extremely varied. Though most studies use single indicators (Fitzgerald 2018), some employ indexes composed of multiple (Borwein and Lucas 2021; John, Fieldhouse, and Liu 2011; Kal Munis 2021; Wong 2010).

over larger ones (Frandsen 2002; Oliver 2000). As Dahl and Tufte noted in their famous study of the topic, this is likely both comes from higher social contact, that residents feel like they can make more of a difference in smaller places, and that they have greater sense of solidarity with fellow residents (Dahl and Tufte 1973). Much of this research studies subnational elections (Frandsen 2002; Lappie and Marschall 2018).

What behaviours we focus on when we study participation is revealing; those studies which look at participation often tend to focus only on voting. They also usually link sub-national geographic identities to local activism, or single issues, particularly on the right (Frandsen 2002; Parry, Moyser, and Day 1992). Far right voting (Cramer 2012; Cramer 2016; Fitzgerald 2018) or resentment or discrimination towards out-groups (Lyons and Utych 2021; Munis 2022) are particularly popular focuses. What this research finds, however, is that there is marked heterogeneity between activities and how they are influenced by geographic identity. Those activities which require more time to complete, or which are otherwise more socially involving, are notably affected. While Davidson and Cotter (1989), for example, find an association between sense of community and voting, contacting politicians and collective action, they find no relation to campaigning or talking to others about politics. These latter two are also actions involving potential conflict or meeting other social groups. Anoll (2018) similarly finds heterogeneity between voting and attending rallies.

There are several issues with this research. Many studies which directly look at the relationship between attachment to place and participation, not just confined to political engagement, use limited samples or do not study variation on small units. The latter is particularly surprising. If, as many do, attempt to construct a theory of attachment to place and its effect on participation primarily as the contextual interaction of different social groups, then why neglect the smallest units, on which social interaction is most frequent, recognisable and direct? This as well as when we know how salient an identity this is. This is often because of lack of good data, though it is also a miss in regards to theory. Secondly, studies may also focus on a specific named identity rather than a general attachment to a geographic unit, particularly when it comes to municipal identity. While this may be helpful—often the more distinctive a place, the stronger the attachment (Lappie and Marschall 2018; Lewicka 2011; Tuan 1975), and voters may find it easier to associate with such named identities—however, it necessarily precludes, because of heterogeneity of interpretation, a study of smaller units, and limits theory to an understanding of that specific identity. [2] A final

[2] Urban sociology studies often centre on the neighbourhood as the most viable unit for political participation (Hays and Kogl 2007; Rohe and Gates 1985).

limitation is often poor attempts at causal identification, in which studies often use small samples, cross-sectional surveys or geographically limited studies. Borwein and Lucas (2021), for example, look at a Canadian city and province; Lappie and Marschall (2018) only focus on mayoral elections in two US states. Hays and Kogl (2007) and Davidson and Cotter (1989) study attachment to place in a handful of participants in Waterloo, Iowa, and Birmingham, Alabama respectively.

The literature broadly suggests two interlinked explanations for why we might expect place attachment to cause us to engage more in political activities: one of strategy, and one of social identity. On the first, it's logical that we would want to defend the places we are attached to. Neighbourhoods are where we spend much of our waking, and increasingly working, lives and there is therefore a selfish motivation for this—we want to live better lives—as well as wanting to see more resources directed to an area we are attached to and therefore care about. Parry et al., argue that this attachment to locality—the people who live there and the values we ascribe to it and them—'encourage action to support and defend its interests and values' (1992, page 300). One of the most prominent ways we may choose to do this is probably through political action. National elections are often cast in terms of the effect of national policies on individual groups, including local areas; in constituency-based electoral systems, there may also be an explicit discussion of local policies and issues and their relation to national politics. While this may be about allocation of resources, there may not even need to be an explicit trade-off in resources discussed. We also know from Chapter 3 that being attached to a place is associated with a lot of behaviours and ways of living which might cause us to defend them through political action—long residences, strong and geographically dense family networks, and homeownership.

Closely related to this is the second explanation: for years, social psychologists have identified the ways in which we place ourselves and others in groups, and assign emotional significance to them (Allport 1954; Tajfel 1981; Tajfel and Turner 1979; Turner 1981). Social identity theory itself has rapidly become popular in political science, as researchers attempt to discern how social identity affects political behaviour, preferences and voting patterns (Mason 2018; Sides, Tesler, and Vavreck 2018). One of the strongest findings from these studies is that strong group identity heightens our knowledge of our group's status in relation to others, from which we derive self-esteem. In doing so, this increases the salience of competition between groups. In as much as elections involve a contest between different social groups, this heightened salience of group identity will trigger behaviour in support of that identity. It is the affective link to the neighbourhood which is key here, however. It is not just defence of your social group when made aware of competition with the out-group, as with the strategic motivation, but a belief in the obligation of engaging in action seen to be in your group's interest, the increased self-worth associated with action which supports your in-group, and the belief that your needs are shared by other members of the group. If we are close to a group, then we also want to engage in activity which we associate with it, which may be perceived to be a group norm, or which we think is in its best interests. The strategic explanation also has an explicitly self-interested motivation, which is lacking here.

Research in environmental or social psychology on different definitions of geographic identity typically finds a robust link with participation in non-political activities. These are wide-ranging, from discussion of and work on, community issues (Bolland and McCallum 2002; Hays and Kogl 2007), to participation in neighbourhood improvement programmes (Lelieveldt 2004). It is relatively easy to imagine this effect on non-political behaviour to carry over into political action, perhaps even national political action. This would require voters to make a connection between political action and identity, however, which is not necessarily as easy to make as with participation in neighbourhood community activities. It would therefore also require them to make a connection specifically to benefits for their area, or support for their social group. Indeed, there is some evidence for this political link. Hays and Kogl (2007), for example, link to voting and attending meetings. Davidson and Cotte (1989) find that sense of community predicts several political activities: voting, contacting officials, and working together on public problems.

> **H_1** Place attachment increases political activity; this does not extend to activities which are too resource or time intensive

A cause for concern with such a theory is that the effect of place attachment on participation may be confounded with social networks, and particularly formal social groups. This is because it would indicate that attachment had no effect on participation directly, through social identity or strategy, and was instead driven merely by network effects and people seeking out social relations or group memberships, particularly those related to political activity. For example, if we are attached to our neighbourhood and this causes us to participate more in a residents' association this, as a result, may drive participation because it is a forum for discussion and engagement. Hays and Kogl (2007), for example, point out that political participation at the neighbourhood level may be driven significantly by membership of formal associations.

Partisan congruence

So far, this is a story in which our affective attachment to place shapes our political activity through our strength of connection to the social groups in that place, and a desire to defend and channel resources there. An important dimension along which this must vary, however, are the actual social groups in the neighbourhood. One way to capture this would be, as many others do, to measure the social or political context of the neighbourhood. This is particularly the case for research which studies the relationship between affective attachment to place and social mix (Bailey, Kearns, and Livingston 2012; Bühlmann 2012; Górny and Toruńczyk-Ruiz 2014; Putnam 2007; Woolever 1992). This however, instead of capturing our relationship with social groups, captures only the social makeup of the place. More apposite would be to study the relationship between the individual and these contextual factors; I did this for race and class in Chapter 3. Measuring it in this way is not new, although on a neighbourhood-level it is more unusual. Van Ham and Feijten (2008), for instance, use a 2002 survey of residents in the Netherlands to assess the role of individual characteristics as they relate to the neighbourhood, focusing on race, and their effect on residents' desire to leave. Oliver (2010) studies the relation of individuals' race to that of their social networks in the US. Laurence and Bentley (2015) measure it as the likelihood that two people in a community will be from the same ethnic group.

In considering political participation, a key element of this relationship is how it varies by the partisan groups which make up the neighbourhood. I suggest that being place-attached pushes us to engage in political action through two mechanisms:

a primarily self-motivated strategic desire to defend the neighbourhood and to see resources directed there and to us; and a social identity explanation, in which our similarity to our in-group and knowledge of its relation to an out-group might cause us to defend it and participate in its visible social norms. Most important to consider here is the role of out-groups. Take the strategic explanation first: If we live in a place and are emotionally attached to it, then it follows that, if we see it as under threat from an out-group we are motivated to defend it. While the strategic motivation is primarily self-interested, we are still members of social groups and therefore aware of out-groups and their potential threat. It is also more explicitly resource-focused, which can be more salient in electoral contexts. If we see a place as potentially under threat by people we consider to be in competition with us, then it may cause us to want to channel resources there, even if it is primarily for our own benefit.

The social identity explanation makes most sense in this regard, however. Firstly, the presence of out-groups may simply make us more aware of them, and therefore trigger action to defend the in-group. It is a common finding in social psychology that out-groups are seen to be more homogeneous than our in-group (Oakes, Haslam, and Turner 1994). Secondly, social identity theory has long established that in-groups become more cohesive and group identity more salient in the presence of out-groups (Tajfel and Turner 1979). If a neighbourhood has people who share a social identity, and if that identity is made salient by the presence of out-groups, then they are likely more motivated to engage in politics in support of that identity. This might happen both in cases where a single out-group is dominant—and therefore is more apparent the more dominant that group is—and where there are many groups, and thus the salience of intergroup competition is high. For example, the literature on spatial economic inequality suggests that inequality within units, as opposed to between, should increase participation, due to increased out-group interaction and elite-led initiatives (Bartle, Birch, and Skirmuntt 2017). Finally, norms regarding participation are also stronger where there is high group cohesion (Larson and Lewis 2017; Oliver 2000).

It helps to consider why inversely this might not be the case. If we are surrounded by members of our in-group, we may be drawn into a false sense of security regarding intergroup competition, and not have our group identity activated. While this generates strong affect towards in-group members (Unger and Wandersman 1985), it means we may not feel we need to defend them through political action. Intergroup conflict in social identity theory is founded on arbitrary group distinctions, often visual, rather than a realistic appraisal of competition with other groups for resources or otherwise. If we are surrounded by fellow partisans, we might not be as aware of other partisans in the neighbourhood and therefore not connect our attachment to the neighbourhood to the need to vote. A core tenet of social identity theory is also social comparison between groups, when intergroup competition is salient (Huddy 2001; Tajfel and Turner 1979). If we do not engage in social comparison because we are unaware of the scale or the potential threat posed by out-groups, then we are unlikely to take up political action because of it.

H_{2a} The effect of place attachment on political participation is stronger in politically incongruent neighbourhoods

H_{2b} This relationship becomes stronger the more politically homogeneous the neighbourhood is

Data and measures

I begin by studying five activities using Round 6 of the European Social Survey (ESS), conducted in 2012 (Sikt 2012). I look in more depth at a smaller selection of activities using three waves of the UK Household Longitudinal Study (UKHLS), covering 2012 to 2018, testing vote propensity in general elections and the 2016 EU referendum, and party behaviour (ISER, NatCen, and Kantar 2021). I first study this cross-sectionally, using wave 9, then use the full panel. This provides a measure of continuity with the previous chapter, as well as allowing us again to use the UKHLS' underutilised high-quality panel and rich selection of variables. A further benefit is the availability of contextual data on a sufficient level of granularity in the UK. However, it is limited beyond voting behaviour and membership.

I first test these conclusions using ESS data, which provide a wider range of specific participation behaviours. I use five of them. The first is voting: whether participants voted in the last national election. I also test engagement with representatives: whether respondents have 'contacted a politician, government or local government official' in the past 12 months. The final three concern more visual or active participation in politics: whether they have signed a petition; 'worn or displayed a campaign badge or materials'; or 'taken part in a lawful public demonstration'; all in the past 12 months. I model on the district level (NUTS-3); as the ESS does not provide districts for all the countries in the study, I instead model only the countries where this is available; see Table A1.

From the UKHLS, I use three variables. The first concerns party-related activities: if respondents are party members, or if they take part in party activities 'on a regular basis', whether they are members or not. The other two ask respondents about voting behaviour: whether they voted in the last general election (2017), and if they voted in the 2016 EU referendum [3] I exclude respondents who were unable to vote. The

[3] Due to the long sampling times for waves in the UKHLS—around 30 months—and the unpredictability of the timing of national elections in the UK, some responses had to be imputed from waves 8 (2017) and 10 (2018).

full questions are on page 47 of Appendix III. While I would have preferred to study the effect of place attachment on voting in local elections rather than national, to my knowledge there is no available survey with this kind of information available, and certainly no panel study which is sufficiently rich.

Descriptive evidence for these variables (Figure 1) shows that those respondents who are place-attached are more likely to engage in all activities and be interested in politics than those who are not, though not to any great degree. In some cases—supporting a political party and being likely to vote in the next election—the difference is more marked. The main predictor is the same as the outcome measure in Chapter 3—the question asking respondents how strongly they agree or disagree with the statement 'I feel like I belong to this neighbourhood' on a five-point Likert scale. I operationalise this as a binary indicator again. I model on the same small geographies as Chapter 3—lower-layer super output areas (LSOAs)—which gives us extremely fine-grained neighbourhood units, to which it is relatively easy to match contextual data.

I also test the proposed mechanism. On strategic motivations, I measure how strongly respondents would be willing to work with others to improve the neighbourhood, and how likely they are to stay. Both of these capture a desire to defend and remain in an area. For social identity explanations, I measure similarity to others in the social group: whether respondents agree that they are similar to their neighbours;

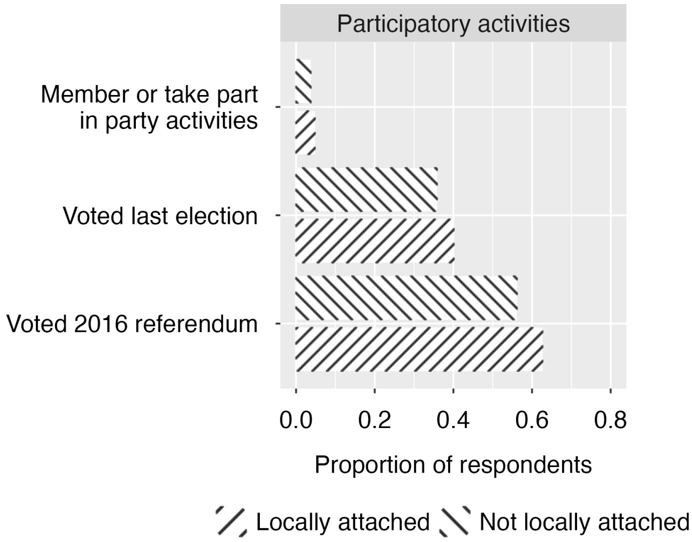

Figure 1: Proportion of respondents engaged in different political activities, comparing those place-attached and those not, including don't knows (UKHLS wave 9)

and whether political participation is seen as a social norm: if they think most of their neighbours vote. Finally, I also measure social networks, both formal group membership and informal social interactions. For the former, I construct a linear additive variable from the question based on whether respondents join in with the activities of groups like unions, tenants associations, religious groups and so on. For the latter, I measure two things: spontaneous social contact in the neighbourhood—whether respondents regularly stop and talk with people—and the proportion of close friends respondents say live in the local area. I also measure affective elements if neighbourhood social capital: general neighbourhood trust, and reciprocity with neighbours. [4]

Measuring partisan congruence

I study variation of this effect by neighbourhood political congruence, using local election partisan vote share. To measure partisan identity, I use elections to choose representatives to municipal authorities, known as 'local authorities' or 'councils'. These authorities have varying powers depending on the region of the country, but typically have control over education, sanitation, planning, and other municipal services. Voters choose candidates to be councillors in their local 'wards'—these have a mean population of 7,490 in 2018, similar in size to LSOA—to fill between one and four seats. [5] I take a collation of all local election results by ward, including by-elections, for the four years prior to the start of the survey (2013–2016), as well as the two survey years: 2017 and 2018 (Teale 2023), and disaggregate to LSOA, matching respondents to the

[4] Due to availability, the former two are matched from Wave 6; there are no subsequent waves in which the questions were asked.

[5] Minimum 158, maximum 37,075, standard deviation 4975) Elections are on a four-yearly cycle: in some authorities all seats are elected at the same time, or they are staggered, typically in thirds. Two electoral systems are used: England and Wales use multi-member plurality voting; Scotland uses the single transferable vote.

closest prior election in their LSOA to their interview date. For details see page 50. [6] To create a measure of political congruence, I use a question asking respondents which party they support or which they are otherwise closer to, with all parties available. 55 per cent of respondents have a party identity. [7] While ideally I would have matched local election data to local election voting preferences, there was no question available for this. I do not study general election vote share for two reasons: vote share for geographic units below the constituency level is not consistently available for general elections in the UK; nor is it possible to reliably estimate vote counts by extrapolating from higher geographies.

I model three congruence thresholds: the plurality party in the neighbourhood, the majority party; and if there is a party winning 60 of the vote. A respondent matches if their identity matches that of the relevant group in their neighbourhood; does not if it does not. Electoral microdata is difficult to obtain in the UK, and the measure I use here represents the most comprehensive data on small geographies available. Most similar studies either use vote share projections, or a sample of the real-world data.

[6] Since I am disaggregating equally to the neighbourhood level from the ward level, I will not be modelling any unique neighbourhood variation in vote share for the partisan congruence, though since these variables are also a reflection of participants' partisanship, they will capture some of that variation. There will, of course, be variation from other neighbourhood-level variables and between-neighbourhood variation captured by the random intercepts at neighbourhood level.

[7] Note, there is no way to separate these two questions.

We might also expect local elections, rather than national, to be most indicative of partisan dynamics shaped by local identities. The overwhelming majority of elections around the world are for sub-national office; these elections are also, in many ways, the most important. They choose the people and government who will be our most immediate contact with representative democracy, the ones who, in most cases, will control roads, waste, sewage and the other essential banalities of municipal life. They choose the people who hold sway over education, transportation and housing, and they may also be the elections which, through planning control, most shape the physical realm.

For both the UKHLS and ESS, I condition on age, gender, education, class and political interest. For the UKHLS, I also condition on race; along with class these are two of the most geographically segregated sociodemographic characteristics in the UK. Age is composed of four binaries: 18 to 24, 25 to 39, 40 to 64, and older than 65. The latter is the reference. Education is three categories: no qualifications, school leavers and those with vocational qualifications, and the university-educated; the first is the reference. Class uses the same three-class schema as I have used before: higher managerial, administrative and professional occupations ('managers', or middle class); intermediate occupations and the self-employed ('intermediate'); and routine and manual occupations ('routine', or working class), the reference category.

[8] Because likelihood to participate is significantly affected by interest in the political process or attention paid to it, I condition on political interest. I operationalise this as a binary: 'High' and 'Low' interest; the latter is the reference. Race for the UKHLS is a simple binary: non-white and white (the reference).

To render this, I primarily use random effects models, fitting random intercepts for groups. For the UKHLS, I initially specify cross-sectional models with a two-level structure, with respondents nested in neighbourhoods. Due to issues with convergence, the party activity models are fixed effects models. [9] For the ESS, I specify a two-level structure, with districts nested within countries, and fit random intercepts for groups, using the same optimiser settings as for the UKHLS models. Some of the congruence models presented too great a challenge for spatially clustered models, with significant convergence issues, and so I specify instead simple logit models, although I re-run the main congruence analysis using a random effects model. For the panel models, I use three waves of the survey—waves 3 (2012), 6 (2015) and 9—specifying an asynchronous model, regressing the outcome in the current period on all values which can vary from the previous period, excluding age, and including the lagged outcome

[8] These are again that the of head of the household—'Household Reference Person'.
[9] Other issues with fit also dictated some choices in specification, particularly changing convergence optimiser and simplifying the structure of the random effects. See page 51.

as a regressor. [10] Unfortunately, because the UKHLS does not collect responses in a wave for all respondents concurrently, I am forced to replace voting in the previous general election with likelihood to vote in the next general election, measuring more likely than not that a respondent will vote. [11] It nonetheless gives us more confidence in the direction of effect.

Results

I start with the basic question of what kind of participatory behaviours place attachment predicts. Table 1 presents five random effects models using the ESS data, looking at voting in the last national election, contacting a politician, wearing a campaign sticker, signing a petition, and protesting. We can see that place attachment positively predicts voting in national elections, and weakly predicts contacting representatives: someone who is place-attached is 6 per cent more likely to have voted in an election, and 2 per cent more likely to have contacted a politician. For wearing

[10] In this case, since the models feature mostly static sociodemographic variables, the only one which can reasonably vary from wave to wave is place attachment.

[11] For waves 2 and 6, this had to be matched from waves 3 and 7 respectively, due to data availability.

campaign materials, signing a petition and protesting, however, the effects are not significant.

Table 1: Voting and other activities (ESS round 6)

	Vote election	Contact politician	Wear materials	Sign petition	Protest
Place attachment	0.32 (0.04)***	0.13 (0.05)**	0.12 (0.06)	0.05 (0.04)	0.06 (0.07)
Political interest	0.85 (0.03)***	0.63 (0.03)***	0.54 (0.04)***	0.48 (0.03)***	0.60 (0.04)***
(Intercept)	–0.76 (0.18)***	–4.21 (0.22)***	–5.20 (0.39)***	–4.03 (0.34)***	–5.41 (0.31)***
Controls	Y	Y	Y	Y	Y
Log Likelihood	–8579.02	–6449.09	–4313.13	–7467.38	–3552.49
N (Individuals)	18062	18714	18712	18689	18707
N (Districts)	147	147	147	147	147
N (Countries)	11	11	11	11	11

Standard errors in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

We want to know why this might be happening, however. Using wave 9 (2018) of the UKHLS I test the two mechanisms on voting in general elections. Initially, we can see that, as with the EVS, place attachment positively predicts vote likelihood. It also does for voting in the 2016 referendum (Table A7), though it is generally weaker, but does not for party activity (Tables A7 and A8 in Appendix III). Someone who is place-attached is 6 per cent more likely to have voted in the general election, the same as for the EVS. Both the strategic and social identity explanations also go some way to explaining this. For voting, for the former, the addition of measures of commitment to the neighbourhood and plans to stay causes the likelihood that someone who is place-attached voted in the election to fall to 4 per cent, from 6; adding those of similarity to others in the neighbourhood and perception of participatory norms causes it also to fall to 4 per cent. In all cases therefore, the strategic and social identity explanations can account for part of the relationship between place attachment and voting, though the overall relationship remains significant.

Table 2: Voting in elections (UKHLS wave 9)

DV: Voted 2017	Model 1	Model 2	Model 3	Model 4
Place attachment	0.47 (0.12)***	0.23 (0.10)*	0.26 (0.10)**	0.36 (0.09)***
Willing to improve area		0.25 (0.06)***		
Plan to stay		0.06 (0.05)		
Similar to others			0.03 (0.05)	
Voting as social norm			0.49 (0.06)***	
High pol interest				1.86 (0.10)***
(Intercept)	5.29 (0.30)***	1.11 (0.36)**	0.13 (0.31)	1.03 (0.17)***
Controls	Y	Y	Y	Y
Log Likelihood	–3594.45	–3607.65	–3487.11	–3392.53
N (Individuals)	9019	9004	8810	8995
N (Neighbourhoods)	4536	4530	4469	4530

Standard errors in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Variation by congruence

How does the relationship between place attachment and participation vary by our exposure to political groups though? Table 3 looks at voting using partisan congruence, using the UKHLS data. When we split the sample, we see that the relationship between place attachment and voting is significant in neighbourhoods which do not match respondents' partisan identity; it is inconsistent or insignificant in those which do (Table ??). In neighbourhoods in which the plurality party is not the same as respondents', someone who is place-attached is 6 per cent more likely to have voted in the general election, compared to someone who is not. Furthermore, this strengthens slightly in increasingly incongruent neighbourhoods. In neighbourhoods which do not match respondents' partisan identity and in which a party won a majority or 60 per cent of the vote, respondents are 7 per cent and 8 per cent more likely to have voted in the general election, respectively. The magnitude of the coefficients here are broadly the same as for the overall model which, while it does not show that the effect is stronger here, does demonstrate that the relationship comes primarily from this group. That the coefficient for place attachment is smaller and not significant in neighbourhoods which match how respondents vote indicates that the relationship is weaker or non existent in this subgroup, and that the overall effect of attachment on voting is dependent on political incongruence: people living in areas which don't vote like them.

Table 3: Election voting congruence: non-matching (UKHLS wave 9)

DV: Voted 2017	Plurality	Majority	60 pc	70 pc
Place attach	0.16 (0.12)	0.17 (0.11)	0.20 (0.11)	0.27 (0.11)*
High pol interest	1.48 (0.17)***	1.57 (0.17)***	1.51 (0.16)***	1.45 (0.15)***
(Intercept)	−0.17 (0.22)	0.06 (0.22)	0.09 (0.21)	0.14 (0.20)
Controls	Y	Y	Y	Y
Log Likelihood	−1420.22	−1619.99	−1742.22	−1831.83
N (Individuals)	2291	2681	2902	3061
N (Neighb'rh'ds)	1703	1936	2056	2142

Standard errors in parentheses.
***p < 0.001; **p < 0.01; *p < 0.05

UPDATE? The null results results for matching neighbourhoods may partly be a product of the smaller sample of neighbourhoods which match respondents' partisan identity in the data, although we observe the same non-significance for referendum voting—which also reflect the election, voting results more generally—where the number of matched and non-matched neighbourhoods is closer in size. [12] For incongruent majority neighbourhoods, someone who is place-attached is 3 per cent more likely to have voted in the referendum, compared to someone who is not. This rises to INSERT for neighbourhoods where one side won at least 60 per cent of the vote.

USE? OR SAY SOMETHING ELSE For referendum voting, the coefficient for place attachment is actually weaker in the non-match model than the overall model, though still significant, while it is not significant in the match model. This result, though weaker, nonetheless still demonstrates significant heterogeneity between the two groups and suggests that place attachment has a consistently measurable impact on participation in the non-matching group.

UPDATE? I also look at several alternative specifications. Because I am modelling on only a subset of the data for the main models, I extend the analysis to all available countries, matched to the EVS sample—EU 27, plus EFTA and the UK—and grouped on countries since NUTS-3 districts are not available for all (Table A5). I also add urbanisation (Table A4), using the same ONS data as Chapter 3. [13] The results in all these cases are substantively the same. For the main UKHLS model, I add

[12] Here, 62 per cent of respondents have a Brexit identity, which had to be matched from a previous wave prior to the referendum, due to data availability. I use two? FINISH congruence thresholds, based on an LSOA-level projection of the vote (Hanretty 2017)—where one side wins a majority, and 60 per cent of the vote.

[13] In this case, I combine 'Large urban' and 'urban'.

urbanisation (Table A10). I also add household as a grouping factor (Table A9); I also re-run the main models for election and referendum voting with fixed effects for neighbourhoods (Table A11). Because size of electoral unit may be a confounding factor, I control for rurality (Table A10). In all these variations, none of the main results change substantively.

UPDATE? One possibility with this question was whether the relationship between place attachment and participation could partly be explained by social networks. The addition of formal group membership and informal contact in the neighbourhood did not have any substantive effect on the relationship between attachment and general election or referendum voting (Table A12), though the strength of place attachment on participation was reduced slightly. The addition of aspects of social capital—generalised trust and reciprocity (Table **??**)—had more of an effect. For both general election and referendum voting, the coefficients for place attachment are no longer significant. I also re-run the subgroup analyses using twp different definitions of congruence. In the first, I use a different question for partisan affiliation which attempts to exclude the forced lean prompt: asking respondents who they would vote for in a hypothetical general election (Table **??**). [14] In the second, I count party proportions

[14] There is no way to actually separate the pure party-ID and lean questions in the UKHLS.

excluding independent candidates (Table A15). The results are FINISH - REMOVE bit on social capital and dont put in results

UPDATE? Finally, I specify a dynamic panel model (Table 4). This is not an ideal specification: because of the UKHLS' very long fieldwork, I cannot use actual behaviour and are instead modelling self-reported likelihood to vote. For similar reasons, I was unable to test voting in the 2016 referendum. There are two reasons that this is not a significant issue, however. The first is that people's intention to vote is generally correlated with their likelihood to actually vote (Glaser 1958; Greenwald et al. 1987). Secondly, these results reflect what we see in the cross-sectional model, even given this relatively hard test of the theory, and therefore should primarily serve to strengthen our conclusions about those. I find that the strength of place attachment on vote propensity is notably reduced from the basic attachment model, though these are not directly comparable. This is perhaps expected, given the specification, but it points to dynamic effects for modelling place attachment on participation. [15]

[15] I also extend the random effects model to the full panel (Table **??**), with random intercepts for individuals, clustering by households nested in neighbourhoods, and separate random intercepts for waves. The results here are similar to the cross-sectional findings

Table 4: Election voting asynchronous panel model (UKHLS)

	Model 1
DV: Voted last election	
Vote propensity (lagged)	2.51 (0.07)***
Place attachment (lagged)	0.20 (0.07)**
(Intercept)	–0.31 (0.13)*
Controls (current period)	Y
Log Likelihood	–2944.55
Deviance	5889.11
N (Individuals)	9024

Standard errors in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Discussion

We know already much about how group identity motivates political behaviour (Conover 1984; Cramer 2016; Huddy 2013; Huddy, Mason, and Aarøe 2015; Miller et al. 1981). We also know that attachment to place, not just the character of the place itself, shapes how we engage in politics (Borwein and Lucas 2021; Bühlmann 2012; Fitzgerald 2018; Lappie and Marschall 2018; Lin and Lunz Trujillo 2022; Wong 2010). What we have now is a better understanding of this link, on a hyperlocal level, and across time. We see this behaviour primarily in low-cost activities, primarily voting, over more resource and time-intensive activities, such as campaigning or protesting. We also know how this can vary by the ways we relate to the political make-up of these places: that the relationship appears driven more by voters living in neighbourhoods which do not look like them politically than by those living in place which do. This effect is also slightly scalable: voters are increasingly motivated to participate in the presence of out-groups of increasing size.

We also know more about why this might be happening. Both strategic and social identity motivations partly account for the relationship between place attachment and voting. Seeing voting in the local area as a social norm—thinking that most people around you do the same—explains most of the relationship in particular. This lends distinct weight to the social identity explanation: that, when we feel as though we belong to our neighbourhood, we engage in these activities more because we see others around us doing so and want to join in, than necessarily any more rational explanation. Voting during national elections is also a relatively high-profile activity and therefore doing it because we want to be seen to do it is probably influential. That this line of reasoning does not therefore carry over into highly-visible activities like campaigning is probably because they are simply too much of an investment for most ordinary voters to bother. There is evidence for the strategic explanation too: wanting to see your neighbourhood improved can explain some of the relationship. That the ability of place attachment to predict voting in the 2016 referendum is consistently weaker than that for voting in elections, gives weight here, since there is a less obvious justification for defending local interests when voting in that case.

I find weak or insignificant relationships between place attachment and more resource-intensive activities, however. If we are to place weight on the strategic explanation, then it may be that voters do not make the connection between other forms of political activity and benefits for their area as they do for voting. That the relationship doesn't extend to wearing a campaign sticker, which is both low-effort and highly visible, would support this. If lean more on the social identity explanation, then it may be that people simply aren't aware enough of the participatory activities of those around them, beyond voting, that they are unable to draw a link between identity and behaviour. This last explanation may also be because the numbers participating in these non-voting activities in both surveys I use are very low.

On congruence, I find that place attachment's ability to predict likelihood to vote is driven by voters living in neighbourhoods which do not look like them, with large numbers of partisan out-groups. I suggested that this might be due primarily by the social identity mechanism: that the presence of out-groups triggers action to defend our in-group, either when we are surrounded by many members of an out-group, or where contestation between different partisan out-groups is high. While the results are not fully consistent, they are broadly born out across a range of models. That we see no significant effect in neighbourhoods dominated by voters' partisan in-group would also somewhat bear this out. As with the main effects, these coefficients are not enormous, though they hold across different specifications. As with other complex social identities—and this is something we may discover as we progress through this work—that its effects are indefinite, and hard to pin down, particularly with the limitations of data.

We knew already from Chapter 3 that place attachment is an identity deeply embedded within many societies, even more strongly held in some cases than attachment to country. This cued us to its potential importance within politics; it seemed unlikely that an identity which is so strongly held would not have some influence on this sphere. I have found that this is the case. This conclusion also sits within a growing collection of studies, dating back many decades, on the role of affective attachment to our environment in shaping participatory behaviours. This chapter has taught us two things: First, that place attachment is related to voting on a very local level, though not more involved political action. Secondly, that, while we already know more about the contextual effect of social mix on political participation, we knew less about the effect of how individual identity interacts with context. I am only studying simple variation on that here, but it seems that the conclusions from this literature—that social mix generates lower participation—are somewhat reversed here.

Referenced works

Allport, Gordon (1954). *The nature of prejudice*. Cambridge, MA: Addison-Wesley

Anoll, Allison P. (2018). "What Makes a Good Neighbor? Race, Place, and Norms of Political Participation". *American Political Science Review* 112.3, pp. 494–508. DOI: 10.1017/S0003055418000175

Arzheimer, Kai and Jocelyn Evans (2012). "Geolocation and voting: Candidate–voter distance effects on party choice in the 2010 UK general election in England". *Political Geography* 31.5, pp. 301–310. DOI: 10.1016/j.polgeo.2012.04.006

— (2014). "Candidate geolocation and voter choice in the 2013 English County Council elections". *Research & Politics* 1.2, pp. 1–9. DOI: 10.1177/2053168014538769

Bailey, Nick, Ade Kearns, and Mark Livingston (2012). "Place Attachment in Deprived Neighbourhoods: The Impacts of Population Turnover and Social Mix". *Housing Studies* 27.2, pp. 208–231. DOI: 10.1080/02673037.2012.632620

Bartle, John, Sarah Birch, and Mariana Skirmuntt (2017). "The local roots of the participation gap: Inequality and voter turnout". *Electoral Studies* 48.1, pp. 30–44. DOI: 10.1016/j.electstud.2017.05.004

Biggs, Michael and Steven Knauss (2012). "Explaining Membership in the British National Party: A Multilevel Analysis of Contact and Threat". *European Sociological Review* 28.5, pp. 633–646. DOI: 10.1093/esr/jcr031

Blok, Lisanne de and Tom van der Meer (2018). "The puzzling effect of residential neighbourhoods on the vote for the radical right: an individual-level panel study on the mechanisms behind neighbourhood effects on voting for the Dutch Freedom Party, 2010–2013". *Electoral Studies* 53.1, pp. 122–132. DOI: 10.1016/j.electstud.2018.04.003

Bolland, John M. and Debra Moehle McCallum (2002). "Neighboring and Community Mobilization in High-Poverty Inner-City Neighborhoods". *Urban Affairs Review* 38.1, pp. 42–69. DOI: 10.1177/107808702401097790

Borwein, Sophie and Jack Lucas (2021). "Municipal Identity and City Interests". *Political Behavior*. DOI: 10.1007/s11109-021-09735-5

Bühlmann, Marc (2012). "Municipal identity. A multilevel analysis of the determinants of individual attachment to municipalities". *Zeitschrift für Vergleichende Politikwissenschaft* 6.2, pp. 149–175. DOI: 10.1007/s12286-011-0115-0

Butler, David and Donald Stokes (1971). *Political Change in Britain: Forces Shaping Electoral Choice*. New edn. UK: Penguin

Cho, Wendy K T and Thomas J Rudolph (2008). "Emanating Political Participation: Untangling the Spatial Structure Behind Participation". *British Journal of Political Science* 38.2, pp. 273–289. DOI: 10.1017/S0007123408000148

Conover, Pamela Johnston (1984). "The Influence of Group Identifications on Political Perception and Evaluation". *The Journal of Politics* 46.3, pp. 760–785. DOI: 10.2307/2130855

Cox, Kevin R (1969). "The voting decision in a spatial context". *Progress in Geography* 1.1, pp. 81–117

Cramer, Katherine (2012). "Putting Inequality in Its Place: Rural Consciousness and the Power of Perspective". *The American Political Science Review* 106.3, pp. 517–532. DOI: 10.1017/S0003055412000305

— (2016). *The Politics of Resentment: Rural Consciousness in Wisconsin and the Rise of Scott Walker*. Illustrated edn. Chicago: University of Chicago Press

Dahl, Robert Alan and Edward R Tufte (1973). *Size and Democracy*. US: Stanford University Press

Davidson, William B and Patrick R Cotte (1989). "Sense of community and political participation". *Journal of Community Psychology* 17.2, pp. 119–125. DOI: 10.1002/1520-6629(198904)17:2<119::aid-jcop2290170203>3.0.co;2-c

Enos, Ryan D (2017). *The Space between Us: Social Geography and Politics*. UK: Cambridge University Press

Fitzgerald, Jennifer (2018). *Close to Home: Local Ties and Voting Radical Right in Europe*. UK: Cambridge University Press

Frandsen, Annie Gaardsted (2002). "Size and Electoral Participation in Local Elections". *Environment and Planning C, Government & Policy* 20.6, pp. 853–869. DOI: 10.1068/c0228

Gerber, Alan, Donald Green, and Christopher W. Larimer (2008). "Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment". *American Political Science Review* 102.1, pp. 33–48. DOI: 10.1017/S000305540808009X

Gimpel, James, Kimberly Karnes, John McTague, and Shanna Pearson-Merkowitz (2008). "Distance-decay in the political geography of friends-and-neighbors voting". *Political Geography* 27.2, pp. 231–252. DOI: 10.1016/j.polgeo.2007.10.005

Glaser, William A. (1958). "Intention and Voting Turnout". *American Political Science Review* 52.4, pp. 1030–1040. DOI: 10.2307/1951983

Górecki, Maciej A. and Michael Marsh (2014). "A decline of 'friends and neighbours voting' in Ireland? Local candidate effects in the 2011 Irish 'earthquake election'". *Political Geography* 41, pp. 11–20. DOI: 10.1016/j.polgeo.2014.03.003

Górny, Agata and Sabina Toruńczyk-Ruiz (2014). "Neighbourhood Attachment in Ethnically Diverse Areas: The Role of Interethnic Ties". *Urban Studies* 51.5, pp. 1000–1018. DOI: 10.1177/0042098013494418

Greenwald, Anthony G., Catherine G. Carnot, Rebecca Beach, and Barbara Young (1987). "Increasing voting behavior by asking people if they expect to vote". *Journal of Applied Psychology* 72.2, pp. 315–318. DOI: 10.1037/0021-9010.72.2.315

Ham, Maarten van and Peteke Feijten (2008). "Who Wants to Leave the Neighbourhood? The Effect of Being Different from the Neighbourhood Population on Wishes to Move". *Environment & Planning A* 40.5, pp. 1151–1170. DOI: 10.1068/a39179

Hanretty, Chris (2017). *Brexit vote projections by output area*. (Visited on 08/28/2023)

Hays, R Allen and Alexandra M Kogl (2007). "Neighborhood Attachment, Social Capital Building, and Political Participation: A Case Study of Low- and Moderate-Income Residents of Waterloo, Iowa". *Journal of Urban Affairs* 29.2, pp. 181–205. DOI: 10.1111/j.1467-9906.2007.00333.x

Huddy, Leonie (2001). "From Social to Political Identity: A Critical Examination of Social Identity Theory". *Political Psychology* 22.1, pp. 127–156. DOI: 10.1111/0162-895X.00230

— (2013). "From Group Identity to Political Cohesion and Commitment". In: *The Oxford Handbook of Political Psychology*. Ed. by Leonie Huddy, David O. Sears, and Jack S. Levy. Oxford University Press, pp. 737–773. (Visited on 09/23/2023)

Huddy, Leonie, Lilliana Mason, and Lene Aarøe (2015). "Expressive Partisanship: Campaign Involvement, Political Emotion, and Partisan Identity". *The American Political Science Review* 109.1, pp. 1–17. DOI: 10.1017/S0003055414000604

ISER, NatCen, and Kantar (2021). *Understanding Society: Waves 1-12, 2009-2021: Special Licence Access, Census 2011 Rural-Urban settlement Indicators*. URL: doi:%2010.5255/UKDA-SN-7630-10

Jankowski, Michael (2016). "Voting for locals: Voters' information processing strategies in open-list PR systems". *Electoral Studies* 43, pp. 72–84. DOI: 10.1016/j.electstud.2016.06.005

John, Peter, Edward Fieldhouse, and Hanhua Liu (2011). "How Civic is the Civic Culture? Explaining Community Participation Using the 2005 English Citizenship Survey". *Political Studies* 59.2, pp. 230–252. DOI: 10.1111/j.1467-9248.2011.00891.x

Johnston, Ron et al. (2000). "The neighbourhood effect and voting in England and wales: Real or imagined?" *British Elections & Parties Review* 10.1, pp. 47–63. DOI: 10.1080/13689880008413036

Johnston, Ron et al. (2005). "Spatial Scale and the Neighbourhood Effect: Multinomial Models of Voting at Two Recent British General Elections". *British Journal of Political Science* 35.3, pp. 487–514

Jöst, Prisca (2023). "Where Do the Less Affluent Vote? The Effect of Neighbourhood Social Context on Individual Voting Intentions in England". *Political Studies* 71.2, pp. 518–541. DOI: 10.1177/00323217211027480

Kal Munis, B (2021). "Place, candidate roots, and voter preferences in an age of partisan polarization: Observational and experimental evidence". *Political Geography* 85.1, p. 102345. DOI: 10.1016/j.polgeo.2021.102345

Key, V.O. (1949). *Southern Politics in State and Nation*. US: University of Tennessee Press

Lappie, John and Melissa Marschall (2018). "Place and participation in local elections". *Political Geography* 64.1, pp. 33–42. DOI: 10.1016/j.polgeo.2018.02.003

Larson, Jennifer M. and Janet I. Lewis (2017). "Ethnic Networks". *American Journal of Political Science* 61.2, pp. 350–364. DOI: 10.1111/ajps.12282

Laurence, James and Lee Bentley (2015). "Does Ethnic Diversity Have a Negative Effect on Attitudes towards the Community? A Longitudinal Analysis of the Causal Claims within the Ethnic Diversity and Social Cohesion Debate". *European Sociological Review* 32.1, pp. 54–67. DOI: 10.1093/esr/jcv081

Lelieveldt, Herman (2004). "Helping Citizens Help Themselves: Neighborhood Improvement Programs and the Impact of Social Networks, Trust, and Norms on Neighborhood-Oriented Forms of Participation". *Urban Affairs Review* 39.5, pp. 531–551. DOI: 10.1177/1078087404263601

Lewicka, Maria (2011). "Place attachment: How far have we come in the last 40 years?" *Journal of Environmental Psychology* 31.3, pp. 207–230. DOI: 10.1016/j.jenvp.2010.10.001

Lewis-Beck, Michael S and Tom W Rice (1983). "Localism in Presidential Elections: The Home State Advantage". *American Journal of Political Science* 27.3, pp. 548–556. DOI: 10.2307/2110984

Lin, Jennifer and Kristin Lunz Trujillo (2022). "Urban-Rural Differences in Non-Voting Political Behaviors". *Political Research Quarterly*. DOI: 10.1177/10659129221119195

Lyons, Jeffrey and Stephen M Utych (2021). "You're Not From Here!: The Consequences of Urban and Rural Identities". *Political Behavior* 45.1, pp. 75–101. DOI: 10.1007/s11109-021-09680-3

Mackay, Charles (1852). *Extraordinary popular delusions and the madness of crowds*. Second Edition. London: LC Page

Mason, Lilliana (2018). *Uncivil Agreement: How Politics Became Our Identity*. Illustrated edition. US: University of Chicago Press

Maxwell, Rahsaan (2019). "Cosmopolitan Immigration Attitudes in Large European Cities: Contextual or Compositional Effects?" *The American Political Science Review* 113.2, pp. 456–474. DOI: 10.1017/S0003055418000898

McClendon, Gwyneth H. (2014). "Social Esteem and Participation in Contentious Politics: A Field Experiment at an LGBT Pride Rally". *American Journal of Political Science* 58.2, pp. 279–290. DOI: 10.1111/ajps.12076

Mckenzie, Brian D. (2004). "Religious Social Networks, Indirect Mobilization, and African-American Political Participation". *Political Research Quarterly* 57.4, pp. 621–632. DOI: 10.1177/106591290405700410

Miller, Arthur H., Patricia Gurin, Gerald Gurin, and Oksana Malanchuk (1981). "Group Consciousness and Political Participation". *American Journal of Political Science* 25.3, pp. 494–511. DOI: 10.2307/2110816

Munis, B. Kal (2022). "Us Over Here Versus Them Over There... Literally: Measuring Place Resentment in American Politics". *Political Behavior* 44.3, pp. 1057–1078. DOI: 10.1007/s11109-020-09641-2

Oakes, Penelope, Alexander Haslam, and John Turner (1994). *Stereotyping and Social Reality*. Oxford: Blackwell

Oliver, E J (2010). *The Paradoxes of Integration: Race, Neighborhood, and Civic Life in Multiethnic America*. Illustrated edition. Chicago: University of Chicago Press

Oliver, Eric (2000). "City Size and Civic Involvement in Metropolitan America". *The American Political Science Review* 94.2, pp. 361–373. DOI: 10.2307/2586017

Parry, Geraint, George Moyser, and Neil Day (1992). *Political Participation and Democracy in Britain*. Cambridge: Cambridge University Press. DOI: 10.1017/CBO9780511558726

Putnam, Robert D (2007). "E pluribus unum: Diversity and community in the twenty-first century the 2006 Johan skytte prize lecture". *Scandinavian Political Studies* 30.2, pp. 137–174. DOI: 10.1111/j.1467-9477.2007.00176.x

Rohe, William M. and Lauren B. Gates (1985). *Planning with Neighborhoods*. UNC Press Books

Sides, John, Michael Tesler, and Lynn Vavreck (2018). *Identity Crisis: The 2016 Presidential Campaign and the Battle for the Meaning of America*. Illustrated edition. US: Princeton University Press

Sikt (2012). *European Social Survey Round 6*. URL: doi:10.21338/NSD-ESS6-2012.

Tajfel, Henri (1981). *Human Groups and Social Categories: Studies in Social Psychology*. UK: Cambridge University Press

Tajfel, Henri and John Turner (1979). "An integrative theory of intergroup conflict". In: *The Social Psychology of Intergroup Relations*. Ed. by William Austin and Stephen Worchel. US: Brooks/Cole, pp. 33–47

Tatalovich, Raymond (1975). ""Friends and Neighbors" Voting". *The Journal of Politics* 37, pp. 807–814. DOI: 10.2307/2129327

Teale, Andrew (2023). *Local elections archive project*

Tuan, Yi-Fu (1975). "Place: An Experiential Perspective". *Geographical Review* 65.2, pp. 151–165. DOI: 10.2307/213970

Turner, John (1981). "The experimental social psychology of intergroup behaviour". In: *Intergroup Behaviour*. Ed. by John Turner and Howard Giles. Oxford: Blackwell

Unger, Donald G and Abraham Wandersman (1985). "The importance of neighbors: The social, cognitive, and affective components of neighboring". *American Journal of Community Psychology* 13.2, pp. 139–169. DOI: 10.1007/BF00905726

Wijk, Daniël van, Gideon Bolt, and Ron Johnston (2019). "Contextual Effects on Populist Radical Right Support: Consensual Neighbourhood Effects and the Dutch PVV". *European Sociological Review* 35.2, pp. 225–238. DOI: 10.1093/esr/jcy049

Wong, Cara J (2010). *Boundaries of Obligation in American Politics: Geographic, National, and Racial Communities*. Illustrated edition. Cambridge: Cambridge University Press

Woolever, Cynthia (1992). "A Contextual Approach to Neighbourhood Attachment". *Urban Studies* 29.1, pp. 99–116. DOI: 10.1080/00420989220080081

Zollinger, Delia (2024). "Place-based identities and cleavage formation in the knowledge society". *Electoral Studies* 88, p. 102768. DOI: 10.1016/j.electstud.2024.102768

Appendix III

Chapter 4

ESS questions

Place attachment

Question

Please say to what extent you agree or disagree with each of the following statements.

I feel close to the people in my local area.

Responses given on a five-point Likert scale.

Participation

Question

Some people don't vote nowadays for one reason or another. Did you vote in the last [country] national election in [month / year]?

Question

There are different ways of trying to improve things in [country] or help prevent things from going wrong. During the last 12 months, have you done any of the following? Have you...

- Contacted a politician, government or local government official?
- Worn or displayed a campaign badge/sticker?
- Signed a petition?
- Taken part in a lawful public demonstration?

Table A1: ESS counties

Country	Main sample	All country sample
Belgium		x
Bulgaria	x	x
Switzerland		x
Cyprus		x
Czechia	x	x
Germany		x
Denmark		x
Estonia	x	x
Spain		x
Finland	x	x
France		x
UK		x
Hungary	x	x
Ireland	x	x
Iceland	x	x
Italy		x
Lithuania	x	x
Netherlands		x
Norway		x
Poland		x
Portugal		x
Sweden	x	x
Slovenia	x	x
Slovakia	x	x

UKHLS questions

Political participation

Question

Are you currently a member of any of the kinds of organisations on this card? [party]

Question

Whether you are a member or not, do you join in the activities of any of these organisations on a regular basis? [party]

Question

Did you vote in this (past) year's general election?

Partisan identity

Question

Now some questions about your views on politics. Generally speaking do you think of yourself as a supporter of any one political party?

Question

Do you think of yourself as a little closer to one political party than to the others?

Positive responses to either of these questions are routed to the same choice question.

Question

If there were to be a general election tomorrow, which political party do you think you would be most likely to support?

Question

Should the United Kingdom remain a member of the European Union or leave the European Union?

Social networks

Question

Here are some statements about neighbourhoods. Please answer how strongly you agree or disagree with each statement.

I regularly stop and talk with people in my neighbourhood.

Responses given on a five-point Likert scale.

Question

Now some questions about your friends. What proportion of your friends live in your local area?

Respondents respond one of 'All are in the local area', 'More than half', 'About half', 'Less than half', or 'Or none?'.

Question

I am going to read a list of types of organisations. For each, tell me whether you are a member of an organisation of that type.

Respondents are presented with a list of organisation types:

Table A2: Group membership options in UKHLS

Groups
Trade Unions
Environmental group
Parents' / School Association
Tenants' / Residents' Group or Neighbourhood Watch
Religious group or church organisation
Voluntary services group
Pensioners group / organisation
Scouts / Guides organisation
Professional organisation
Other community or civic group
Social Club / Working men's club
Sports Club
Women's Institute / Townswomen's Guild
Women's Group / Feminist Organisation

Strategic explanations

Question

I would be willing to work together with others on something to improve my neighbourhood.

Question

I plan to remain a resident of this neighbourhood for a number of years.

Responses given on a five-point Likert scale.

Social identity explanations

Question

I think of myself as similar to the people that live in this neighbourhood.

Question

Most people around here usually vote in general elections.

Responses given on a five-point Likert scale.

Notes on data and coding for UKHLS and ESS

- Where UK wards are multi-member, I take the candidate with the most votes as that party's share of the vote in that ward, in cases where parties stand multiple candidates. If respondents' partisan identity is missing in the survey, I count this as no partisan match.
- Occasionally councils released results by incorrect ward, typically using older ward boundaries. These I hand code to their correct versions. While LSOAs do not nest perfectly within wards, they are in general very close fits, such that it is possible to match them near uniformly using standard lookup tables.
- Models excluding independent candidates are excluding only truly independent candidates, and not independent candidates who stand on behalf of small 'independent' parties.
- For those not in work, class is coded to their last occupation. Full-time students, and those who have never worked or are involuntarily unemployed for more than a year—are excluded from the sample. This uses approximate social grade for Scotland, due to data unavailability.
- Occupation for the ESS is measured using the 2008 International Standard Occupationification of Occupations (ISCO-08), which I match to the NS-SeC classes. This conversation was performed at the ISOC 'major' level, based principally on the service relationship and labour contract nature of the occupation. Since they use different classification methodologies, there are no 'intermediate' occupations in ISCO-08 at the major level.
-
- For referendum voting in the UKHLS, because the question is not asked for every wave, I matched Brexit vote from waves 10–12 for wave 9. The variable

is unevenly captured, however, since for wave 10 it was only asked in 2019, and for waves 11 and 12 only around the European and general elections of 2019.

Issues with model fit for UKHLS data

Issues with convergence in the main random effects models for Chapter 4 likely was due to the small counts in the outcome being perfectly predicted by some or a combination of the predictors, or causing a singularity in the covariance matrix. To address this, firstly, I combined both party membership and activity into one outcome measure. Secondly, I simplified both the structure of the random effects—omitting the household level—and some of the predictors, though I introduce different specifications in this Appendix. Finally, I specified an optimisation algorithm design to address convergence issues and increasing the maximum number of function evaluations to give the models more opportunity to converge. This is the Bound Optimisation By Quadratic Approximation, which is better suited than other standard optimisers to dealing with complex models or ones which struggle to converge. All other random effects models in this thesis use the Nelder-Mead, more appropriate when there is more variation in the outcome measure.

ESS other models

Table A3: Voting and other activities, full model (ESS round 6)

	Vote election	Contact politician	Wear sticker	Sign petition	Protest
Place attachment	0.32 (0.04)***	0.13 (0.05)**	0.12 (0.06)	0.05 (0.04)	0.06 (0.07)
Age (ref = 65+)					
18 to 24	−0.98 (0.09)***	−0.04 (0.12)	0.81 (0.12)***	0.78 (0.10)***	1.27 (0.15)***
25 to 39	−0.60 (0.06)***	0.21 (0.07)**	0.32 (0.09)***	0.80 (0.07)***	0.79 (0.11)***
40 to 64	−0.14 (0.05)**	0.38 (0.06)***	0.27 (0.08)***	0.57 (0.06)***	0.65 (0.10)***
Female	0.14 (0.04)***	−0.18 (0.05)***	0.40 (0.06)***	0.27 (0.04)***	−0.09 (0.07)
Education (ref = No quals)					
School leaver	0.05 (0.09)	0.20 (0.11)	0.33 (0.14)*	0.30 (0.10)**	−0.03 (0.17)
Degree	0.41 (0.10)***	0.59 (0.12)***	0.52 (0.16)***	0.59 (0.12)***	0.31 (0.19)
Manager	0.24 (0.05)***	0.30 (0.05)***	0.24 (0.07)***	0.27 (0.05)***	0.17 (0.08)*
Political interest	0.85 (0.03)***	0.63 (0.03)***	0.54 (0.04)***	0.48 (0.03)***	0.60 (0.04)***
(Intercept)	−0.76 (0.18)***	−4.21 (0.22)***	−5.20 (0.39)***	−4.03 (0.34)***	−5.41 (0.31)***
Log Likelihood	−8579.02	−6449.09	−4313.13	−7467.38	−3552.49
N (Individuals)	18062	18714	18712	18689	18707
N (Districts)	147	147	147	147	147
N (Countries)	11	11	11	11	11

Standard errors in parentheses.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table A4: Participation models, adding urbanisation (ESS round 6)

	Vote election	Contact politician	Wear sticker	Sign petition	Protest
Place attachment	0.29 (0.04)***	0.12 (0.05)*	0.12 (0.06)	0.05 (0.05)	0.05 (0.07)
Age (ref = 65+)					
18 to 24	–1.01 (0.09)***	–0.10 (0.13)	0.83 (0.12)***	0.76 (0.10)***	1.24 (0.15)***
25 to 39	–0.60 (0.06)***	0.17 (0.08)*	0.32 (0.09)***	0.79 (0.07)***	0.74 (0.11)***
40 to 64	–0.13 (0.05)*	0.37 (0.06)***	0.28 (0.08)***	0.58 (0.06)***	0.63 (0.10)***
Female	0.16 (0.04)***	–0.15 (0.05)**	0.41 (0.06)***	0.27 (0.04)***	–0.08 (0.07)
Education (ref = No quals)					
School leaver	0.03 (0.09)	0.22 (0.11)	0.33 (0.14)*	0.29 (0.10)**	0.00 (0.18)
Degree	0.34 (0.11)**	0.60 (0.12)***	0.51 (0.16)**	0.58 (0.12)***	0.32 (0.19)
Manager	0.25 (0.05)***	0.31 (0.06)***	0.25 (0.07)***	0.26 (0.05)***	0.17 (0.08)*
Political interest	0.88 (0.03)***	0.63 (0.03)***	0.54 (0.04)***	0.47 (0.03)***	0.60 (0.04)***
Urbanisation (ref = Pred. rural)					
Predominantly urban	–0.08 (0.09)	–0.58 (0.12)***	–0.17 (0.15)	0.04 (0.13)	0.36 (0.18)*
Intermediate rural	–0.08 (0.07)	–0.31 (0.10)**	–0.01 (0.12)	–0.01 (0.11)	–0.04 (0.15)
(Intercept)	–0.73 (0.19)***	–3.92 (0.23)***	–5.03 (0.41)***	–3.94 (0.37)***	–5.43 (0.33)***
Log Likelihood	–8065.94	–6164.76	–4229.72	–7190.98	–3392.63
N (Individuals)	17055	17679	17679	17654	17674
N (Districts)	135	135	135	135	135
N (Countries)	10	10	10	10	10

Standard errors in parentheses.
***p < 0.001; **p < 0.01; *p < 0.05

Table A5: All countries in the sample (ESS)

	Vote election	Contact politician	Wear sticker	Sign petition	Protest
Place attachment	0.33 (0.03)***	0.24 (0.03)***	0.22 (0.04)***	0.08 (0.03)**	0.06 (0.04)
Age (ref = 65+)					
18 to 24	–1.05 (0.06)***	–0.08 (0.08)	0.97 (0.08)***	0.70 (0.06)***	1.27 (0.09)***
25 to 39	–0.79 (0.04)***	0.22 (0.05)***	0.44 (0.07)***	0.63 (0.04)***	0.76 (0.07)***
40 to 64	–0.26 (0.04)***	0.37 (0.04)***	0.40 (0.06)***	0.50 (0.04)***	0.63 (0.06)***
Female	0.10 (0.03)***	–0.21 (0.03)***	0.39 (0.04)***	0.27 (0.03)***	–0.07 (0.04)
Education (ref = No quals)					
School leaver	0.05 (0.05)	0.28 (0.07)***	0.34 (0.09)***	0.49 (0.06)***	0.30 (0.09)***
Degree	0.39 (0.06)***	0.54 (0.08)***	0.45 (0.10)***	0.79 (0.07)***	0.56 (0.10)***
Manager	0.36 (0.03)***	0.34 (0.04)***	0.28 (0.05)***	0.31 (0.03)***	0.27 (0.05)***
Political interest	0.74 (0.02)***	0.58 (0.02)***	0.59 (0.03)***	0.47 (0.02)***	0.58 (0.03)***
(Intercept)	–0.37 (0.12)**	–4.13 (0.12)***	–5.44 (0.22)***	–3.88 (0.19)***	–5.39 (0.20)***
Log Likelihood	–17250.08	–14490.95	–9312.89	–18230.98	–8667.27
N (Individuals)	37875	39640	39644	39570	39637
N (Countries)	24	24	24	24	24

Standard errors in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

UKHLS other models

Table A6: Voting in elections, full model (UKHLS wave 9)

DV: Voted 2017	Model 1	Model 2	Model 3	Model 4
Place attachment	0.47 (0.12)***	0.23 (0.10)*	0.26 (0.10)**	0.36 (0.09)***
Age (ref = 65+)				
18 to 24	–1.50 (0.23)***	–1.49 (0.19)***	–1.54 (0.17)***	–1.27 (0.17)***
25 to 39	–1.36 (0.20)***	–1.48 (0.16)***	–1.47 (0.15)***	–1.16 (0.14)***
40 to 64	–0.33 (0.17)	–0.66 (0.13)***	–0.64 (0.12)***	–0.46 (0.12)***
Female	0.12 (0.10)	0.06 (0.08)	0.07 (0.08)	0.30 (0.08)***
Education (ref = No quals)				
School leaver	0.99 (0.20)***	0.66 (0.16)***	0.63 (0.15)***	0.45 (0.15)**
Degree	2.26 (0.23)***	1.78 (0.20)***	1.72 (0.17)***	1.33 (0.17)***
Occupation (ref = Routine)				
Manager	0.61 (0.17)***	0.82 (0.13)***	0.77 (0.12)***	0.61 (0.11)***
Intermediate	0.82 (0.20)***	0.74 (0.15)***	0.67 (0.13)***	0.58 (0.13)***
Non-white	–0.55 (0.22)*	–0.51 (0.15)***	–0.51 (0.14)***	–0.47 (0.13)***
Willing to improve area		0.25 (0.06)***		
Plan to stay		0.06 (0.05)		
Similar to others			0.03 (0.05)	
Voting as social norm			0.49 (0.06)***	
High pol interest				1.86 (0.10)***
(Intercept)	5.29 (0.30)***	1.11 (0.36)**	0.13 (0.31)	1.03 (0.17)***
Log Likelihood	–3594.45	–3607.65	–3487.11	–3392.53
N (Individuals)	9019	9004	8810	8995
N (Neighbourhoods)	4536	4530	4469	4530

Standard errors in parentheses. ***p < 0.001; **p < 0.01; *p < 0.05

Table A7: Party activity (UKHLS wave 9)

DV: Party member or activity	Model 1	Model 2	Model 3	Model 4
Place attachment	0.07 (0.20)	0.01 (0.22)	0.19 (0.22)	0.15 (0.23)
Age (ref = 65+)				
18 to 24	0.08 (0.34)	–0.01 (0.35)	0.03 (0.35)	0.11 (0.41)
25 to 39	–0.60 (0.33)	–0.68 (0.33)*	–0.56 (0.34)	–0.34 (0.40)
40 to 64	–0.66 (0.25)**	–0.71 (0.25)**	–0.59 (0.26)*	–0.78 (0.30)**
Female	–0.67 (0.14)***	–0.71 (0.15)***	–0.66 (0.15)***	–0.38 (0.17)*
Education (ref = No quals)				
School leaver	0.31 (0.47)	0.20 (0.47)	0.24 (0.47)	–0.26 (0.53)
Degree	1.51 (0.48)**	1.37 (0.49)**	1.39 (0.48)**	0.74 (0.54)
Occupation (ref = Routine)				
Manager	0.67 (0.36)	0.63 (0.36)	0.63 (0.37)	0.34 (0.38)
Intermediate	0.74 (0.40)	0.72 (0.40)	0.71 (0.41)	0.47 (0.43)
Non-white	–0.18 (0.34)	–0.17 (0.35)	–0.18 (0.35)	–0.23 (0.40)
Willing to improve area		0.42 (0.13)**		
Plan to stay		–0.18 (0.11)		
Similar to others			–0.16 (0.10)	
Voting as social norm			0.12 (0.13)	
High pol interest				4.12 (0.43)***
Log Likelihood	–954.15	–943.21	–941.30	–785.55
N (Individuals)	1744	1742	1708	1739
N (Neighbourhoods)	503	503	499	501

Standard errors in parentheses. Fixed effects for neighbourhood. ***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Table A8: Voting in EU referendum (UKHLS wave 9)

DV: Voted EU ref	Model 1	Model 2	Model 3	Model 4
Place attachment	0.25 (0.05)***	0.13 (0.06)*	0.19 (0.06)**	0.18 (0.05)**
Age (ref = 65+)				
18 to 24	–1.13 (0.11)***	–1.09 (0.11)***	–1.13 (0.11)***	–0.92 (0.11)***
25 to 39	–0.91 (0.08)***	–0.91 (0.08)***	–0.90 (0.09)***	–0.64 (0.09)***
40 to 64	–0.46 (0.07)***	–0.48 (0.07)***	–0.44 (0.07)***	–0.30 (0.07)***
Female	–0.21 (0.05)***	–0.22 (0.05)***	–0.20 (0.05)***	0.01 (0.05)
Education (ref = No quals)				
School leaver	0.55 (0.09)***	0.51 (0.09)***	0.52 (0.09)***	0.39 (0.09)***
Degree	1.46 (0.10)***	1.40 (0.10)***	1.42 (0.10)***	1.09 (0.10)***
Occupation (ref = Routine)				
Manager	0.76 (0.07)***	0.74 (0.07)***	0.71 (0.07)***	0.60 (0.07)***
Intermediate	0.57 (0.07)***	0.56 (0.07)***	0.54 (0.08)***	0.48 (0.08)***
Non-white	–0.58 (0.08)***	–0.58 (0.08)***	–0.60 (0.08)***	–0.60 (0.08)***
Willing to improve area		0.17 (0.03)***		
Plan to stay		0.02 (0.03)		
Similar to others			0.00 (0.03)	
Voting as social norm			0.27 (0.04)***	
High pol interest				1.54 (0.06)***
(Intercept)	0.95 (0.10)***	0.36 (0.15)*	0.05 (0.18)	0.35 (0.10)***
Log Likelihood	–6839.93	–6809.30	–6638.31	–6411.74
N (Individuals)	14334	14315	14035	14299
N (Neighbourhoods)	7364	7356	7267	7357

Standard errors in parentheses. ***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Table A9: Participation models, adding household as grouping factor (UKHLS wave 9)

	Vote election	Vote EU ref
Place attachment	0.41 (0.16)**	0.30 (0.06)***
Age (ref = 65+)		
18 to 24	–1.08 (0.28)***	–1.33 (0.13)***
25 to 39	–0.90 (0.26)***	–1.03 (0.10)***
40 to 64	0.36 (0.23)	–0.51 (0.08)***
Female	0.25 (0.13)	–0.23 (0.06)***
Education (ref = No quals)		
School leaver	1.02 (0.25)***	0.66 (0.11)***
Degree	2.32 (0.29)***	1.67 (0.12)***
Occupation (ref = Routine)		
Manager	0.56 (0.26)*	0.94 (0.09)***
Intermediate	0.47 (0.30)	0.72 (0.10)***
Non-white	–0.65 (0.28)*	–0.70 (0.09)***
(Intercept)	6.58 (0.30)***	1.14 (0.13)***
Log Likelihood	–3250.89	–6774.66
N (Individuals)	9019	14334
N (Households)	5450	9000
N (Neighbourhoods)	4536	7364

Standard errors in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Table A10: Participation models, adding urbanisation (UKHLS wave 9)

	Vote election	Vote EU ref
Place attachment	0.47 (0.00)***	0.25 (0.05)***
Age (ref = 65+)		
18 to 24	–1.50 (0.21)***	–1.13 (0.11)***
25 to 39	–1.36 (0.18)***	–0.91 (0.08)***
40 to 64	–0.33 (0.16)*	–0.46 (0.07)***
Female	0.12 (0.10)	–0.21 (0.05)***
Education (ref = No quals)		
School leaver	0.98 (0.12)***	0.55 (0.09)***
Degree	2.24 (0.00)***	1.46 (0.10)***
Occupation (ref = Routine)		
Manager	0.61 (0.00)***	0.76 (0.07)***
Intermediate	0.82 (0.18)***	0.56 (0.07)***
Non-white	–0.55 (0.22)*	–0.58 (0.08)***
Urbanisation (ref = Rural)		
Urban	–0.04 (0.17)	–0.13 (0.10)
Suburbs	–0.05 (0.32)	–0.18 (0.13)
(Intercept)	5.34 (0.00)***	1.08 (0.14)***
Log Likelihood	–3594.43	–6838.78
N (Individuals)	9019	14334
N (Neighbourhoods)	4536	7364

Standard errors in parentheses.
***p < 0.001; **p < 0.01; *p < 0.05

Table A11: Participation fixed effects models (UKHLS wave 9)

	Party activity	Vote election	Vote EU ref
Place attachment	0.07 (0.20)	0.38 (0.12)**	0.40 (0.17)*
Age (ref = 65+)			
18 to 24	0.08 (0.34)	–0.81 (0.24)***	–1.01 (0.33)**
25 to 39	–0.60 (0.33)	–0.31 (0.21)	–0.85 (0.29)**
40 to 64	–0.66 (0.25)**	0.29 (0.17)	0.10 (0.25)
Female	–0.67 (0.14)***	–0.27 (0.09)**	0.14 (0.14)
Education (ref = No quals)			
School leaver	0.31 (0.47)	0.57 (0.22)**	1.25 (0.31)***
Degree	1.51 (0.48)**	1.44 (0.24)***	2.34 (0.35)***
Occupation (ref = Routine)			
Manager	0.67 (0.36)	0.81 (0.22)***	0.18 (0.32)
Intermediate	0.74 (0.40)	0.48 (0.22)*	0.86 (0.35)*
Non-white	–0.18 (0.34)	–1.06 (0.29)***	–0.48 (0.41)
(Intercept)	–0.42 (0.22)	–0.30 (0.14)*	–0.32 (0.12)*
Log Likelihood	–954.15	–2386.83	–1146.99
N (Individuals)	1744	4044	2000
N (Neighbourhoods)	503	1294	611

Standard errors in parentheses. Fixed effects for neighbourhood
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Table A12: Participation models, adding social networks (UKHLS wave 9)

	Vote election	Vote election	Vote EU ref	Vote EU ref
Place attachment	0.38 (0.09)***	0.27 (0.10)**	0.20 (0.05)***	0.17 (0.06)**
Age (ref = 65+)				
18 to 24	–1.47 (0.19)***	–1.41 (0.19)***	–1.05 (0.11)***	–1.03 (0.11)***
25 to 39	–1.42 (0.16)***	–1.39 (0.16)***	–0.83 (0.08)***	–0.83 (0.08)***
40 to 64	–0.58 (0.13)***	–0.59 (0.13)***	–0.42 (0.07)***	–0.41 (0.07)***
Female	0.06 (0.08)	0.05 (0.08)	–0.22 (0.05)***	–0.22 (0.05)***
Education (ref = No quals)				
School leaver	0.64 (0.16)***	0.64 (0.16)***	0.49 (0.09)***	0.49 (0.09)***
Degree	1.72 (0.20)***	1.72 (0.20)***	1.33 (0.10)***	1.33 (0.10)***
Occupation (ref = Routine)				
Manager	0.78 (0.13)***	0.78 (0.12)***	0.70 (0.07)***	0.70 (0.07)***
Intermediate	0.72 (0.15)***	0.72 (0.15)***	0.55 (0.07)***	0.55 (0.07)***
Non-white	–0.54 (0.16)***	–0.52 (0.15)***	–0.58 (0.08)***	–0.58 (0.08)***
Organisational membership	0.31 (0.06)***	0.30 (0.06)***	0.27 (0.03)***	0.27 (0.03)***
Neighbour contact		0.11 (0.05)*		0.03 (0.03)
(Intercept)	2.01 (0.34)***	1.64 (0.33)***	0.88 (0.10)***	0.78 (0.13)***
Log Likelihood	–3610.49	–3607.79	–6797.92	–6797.36
N (Individuals)	9016	9016	14329	14329
N (Neighbourhoods)	4535	4535	7363	7363

Standard errors in parentheses. ***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Table A13: Election voting asynchronous panel model, full model (UKHLS)

DV: Voted last election	Model 1
Lagged Election voting	2.51 (0.07)***
Lagged Place attachment	0.20 (0.07)**
Age (ref = 65+)	
18 to 24	–0.56 (0.17)***
25 to 39	–0.89 (0.11)***
40 to 64	–0.62 (0.10)***
Female	–0.02 (0.07)
Education (ref = No quals)	
School leaver	0.42 (0.11)***
Degree	0.96 (0.13)***
Occupation (ref = Routine)	
Manager	0.50 (0.08)***
Intermediate	0.19 (0.09)*
Race (ref = White)	
Black	–0.34 (0.22)
South Asian	0.47 (0.20)*
Other race	–0.27 (0.21)
(Intercept)	–0.31 (0.13)*
Log Likelihood	–2944.55
Deviance	5889.11
N (Individuals)	9024

Standard errors in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

UKHLS congruence subgroup analysis

Table A14: Election voting congruence: non-matching, full model (UKHLS wave 9)

DV: Voted 2017	Plurality	Majority	60 per cent	70 per cent
Place attachment	0.16 (0.12)	0.17 (0.11)	0.20 (0.11)	0.27 (0.11)*
Age (ref = 65+)				
18 to 24	–1.10 (0.24)***	–1.13 (0.23)***	–1.16 (0.22)***	–1.11 (0.21)***
25 to 39	–1.02 (0.20)***	–1.05 (0.19)***	–1.01 (0.18)***	–0.99 (0.18)***
40 to 64	–0.48 (0.17)**	–0.52 (0.17)**	–0.52 (0.16)**	–0.50 (0.16)**
Female	0.34 (0.11)**	0.38 (0.11)***	0.37 (0.11)***	0.35 (0.10)***
Education (ref = No quals)				
School leaver	0.31 (0.20)	0.27 (0.19)	0.34 (0.18)	0.32 (0.18)
Degree	1.03 (0.23)***	1.00 (0.22)***	1.01 (0.21)***	0.96 (0.20)***
Occupation (ref = Routine)				
Manager	0.39 (0.15)**	0.42 (0.14)**	0.40 (0.14)**	0.40 (0.13)**
Intermediate	0.40 (0.17)*	0.43 (0.16)**	0.41 (0.16)**	0.42 (0.15)**
Non-white	–0.56 (0.19)**	–0.58 (0.18)**	–0.33 (0.16)*	–0.17 (0.15)
High pol interest	1.48 (0.17)***	1.57 (0.17)***	1.51 (0.16)***	1.45 (0.15)***
(Intercept)	–0.17 (0.22)	0.06 (0.22)	0.09 (0.21)	0.14 (0.20)
Log Likelihood	–1420.22	–1619.99	–1742.22	–1831.83
N (Individuals)	2291	2681	2902	3061
N (Neighbourhoods)	1703	1936	2056	2142

Standard errors in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Table A15: Election voting congruence: non-matching, not counting independents (UKHLS wave 9)

DV: Voted 2017	Plurality	Majority	60 per cent	70 per cent
Place attachment	0.16 (0.12)	0.15 (0.11)	0.18 (0.11)	0.25 (0.11)*
Age (ref = 65+)				
18 to 24	–1.10 (0.24)***	–1.21 (0.23)***	–1.17 (0.22)***	–1.13 (0.22)***
25 to 39	–1.02 (0.20)***	–1.05 (0.19)***	–1.01 (0.19)***	–1.01 (0.18)***
40 to 64	–0.48 (0.17)**	–0.50 (0.17)**	–0.50 (0.16)**	–0.50 (0.16)**
Female	0.34 (0.11)**	0.37 (0.11)***	0.37 (0.11)***	0.34 (0.10)***
Education (ref = No quals)				
School leaver	0.31 (0.20)	0.29 (0.19)	0.32 (0.18)	0.31 (0.18)
Degree	1.03 (0.23)***	1.03 (0.22)***	1.00 (0.21)***	0.96 (0.21)***
Occupation (ref = Routine)				
Manager	0.39 (0.15)**	0.40 (0.14)**	0.40 (0.14)**	0.41 (0.13)**
Intermediate	0.40 (0.17)*	0.44 (0.17)**	0.44 (0.16)**	0.44 (0.15)**
Non-white	–0.56 (0.19)**	–0.54 (0.18)**	–0.32 (0.16)	–0.17 (0.16)
High pol interest	1.48 (0.17)***	1.56 (0.17)***	1.52 (0.16)***	1.47 (0.15)***
(Intercept)	–0.17 (0.22)	0.02 (0.22)	0.09 (0.21)	0.14 (0.20)
Log Likelihood	–1420.22	–1591.13	–1716.37	–1804.42
N (Individuals)	2291	2625	2851	3015
N (Neighbourhoods)	1703	1904	2031	2123

Standard errors in parentheses.
***p < 0.001; **p < 0.01; *p < 0.05